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Pledge: I pledge my honor that I have abided by the Stevens Honor System.

Lab: Elementary Sorting. Array after each outer loop iteration.

1. Bubble Sort Optimized

Start

0	1	2	3	4	5
78	15	23	2	97	85

Pass 1: 78 bubbles right, then 97 and 85 swap. Swaps made.

0	1	2	3	4	5
15	23	2	78	85	97

Pass 2: 23 and 2 swap. Swaps made.

0	1	2	3	4	5
15	2	23	78	85	97

Pass 3: 15 and 2 swap. Swaps made.

0	1	2	3	4	5
2	15	23	78	85	97

Pass 4: no swaps, so the loop ends.

0	1	2	3	4	5
2	15	23	78	85	97

2. Selection Sort

Start

0	1	2	3	4	5
78	15	23	2	97	85

i = 0: smallest is 2, swap with 78.

0	1	2	3	4	5
2	15	23	78	97	85

i = 1: smallest is 15, already in place.

0	1	2	3	4	5
2	15	23	78	97	85

i = 2: smallest is 23, already in place.

0	1	2	3	4	5
2	15	23	78	97	85

i = 3: smallest is 78, already in place.

0	1	2	3	4	5
2	15	23	78	97	85

i = 4: smallest is 85, swap with 97.

0	1	2	3	4	5
2	15	23	78	85	97

3. Insertion Sort

Start

0	1	2	3	4	5
78	15	23	2	97	85

i = 1: insert 15 before 78.

0	1	2	3	4	5
15	78	23	2	97	85

i = 2: insert 23 before 78.

0	1	2	3	4	5
15	23	78	2	97	85

i = 3: insert 2 at the front.

0	1	2	3	4	5
2	15	23	78	97	85

i = 4: 97 stays, already larger than 78.

0	1	2	3	4	5
2	15	23	78	97	85

i = 5: insert 85 before 97.

0	1	2	3	4	5
2	15	23	78	85	97